

30-Day Python Mastery Plan (Beginner to Pro)

Week 1 – Python Basics

Day 1 – Variables & I/O

- Add, subtract, multiply, divide two numbers
- Convert Celsius to Fahrenheit
- Swap two variables
- Area of circle
- Convert seconds to time format

Day 2 – Conditionals

- Even or odd
- Largest of three numbers
- Leap year check
- Grade calculator
- Positive / Negative / Zero

Day 3 – Loops

- Print 1 to N
- Factorial
- Reverse a number
- Sum of digits
- Prime numbers in range

Day 4 – Strings

- Reverse string
- Palindrome check
- Count vowels
- Character frequency
- Anagram check

Day 5 – Lists

- Largest & smallest
- Remove duplicates
- Second largest
- Sort without built-in
- Rotate list

Day 6 – Tuples & Sets

- List to tuple
- Common elements
- Tuple palindrome
- Remove duplicates
- Union & intersection

Day 7 – Mini Project

- Number guessing game
- Menu-driven calculator

Week 2 – Core Python

Day 8 – Dictionaries

- Word frequency counter
- Max value key
- Sort by values
- Merge dictionaries
- Invert dictionary

Day 9 – Functions

- Prime function
- Recursive factorial
- Fibonacci
- Armstrong number
- Multiple return values

Day 10 – File Handling

- Count lines
- Count words & characters
- Copy file
- Most frequent word
- Remove blank lines

Day 11 – Exception Handling

- Division by zero
- Custom exception
- File not found
- Multiple exceptions
- Safe calculator

Day 12 – Modules & Packages

- Custom math module
- Password generator
- Datetime clock
- Alias import
- Create package

Day 13 – Regular Expressions

- Email validation
- Phone validation
- Extract numbers
- Duplicate words
- Password strength checker

Day 14 – Project

- Contact Book with file storage and error handling

Week 3 – Advanced Python

Day 15 – OOP Basics

- Student class
- Bank Account class
- Constructor usage
- Class vs instance variables
- Object interaction

Day 16 – OOP Advanced

- Inheritance
- Method overriding
- Encapsulation
- Polymorphism
- Library Management System

Day 17 – Data Structures

- Stack
- Queue
- Balanced parentheses
- Reverse stack
- Circular queue

Day 18 – Algorithms

- Linear search
- Binary search
- Bubble sort
- Selection sort
- Merge sort

Day 19 – Multithreading

- Create threads
- Concurrent execution
- Thread-safe counter
- Producer-consumer
- Execution time comparison

Day 20 – Multiprocessing

- Parallel factorial
- CPU vs I/O tasks
- Thread vs process comparison
- Parallel sum
- Process communication

Day 21 – Project

- Command-line To-Do App

Week 4 – Pro Level

Day 22 – APIs

- Consume API
- Parse JSON
- Weather fetcher
- API error handling
- Formatted output

Day 23 – Flask

- Create Flask app
- Routes
- GET & POST
- JSON API
- REST API

Day 24 – Web Scraping

- Scrape headlines
- Extract links
- Store data
- Handle errors
- Clean text

Day 25 – Databases

- Create DB
- Insert data
- Fetch data
- Update records
- Delete records

Day 26 – ORM

- Create models
- Insert using ORM
- Query records
- Update records
- Delete records

Day 27 – System Design

- URL shortener design
- Expense tracker design
- Chat app flow
- Recommendation logic
- API + DB flow

Day 28–30 – Capstone Project

- Expense Tracker / Weather App / Student Management / Chat App / Recommendation System

